Week Ten: Data Visualization for Inventory Management System using JavaScript

Aim

To create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

Procedure

1. Set up the HTML file with two <canvas> elements for pie chart and bar chart.
2. Include **Chart.js** library for creating the charts.
3. Write JavaScript code (script.js) to define inventory data and render both charts.
4. Style the page using basic internal CSS.

Program Code

1. HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Form Validation</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
     <form id="myForm">
       <label for="email">Email:</label>
       <input type="email" id="email" name="email" required>
       <span id="emailError" class="error"></span>

       <label for="phone">Phone Number:</label>
       <input type="text" id="phone" name="phone" required>
       <span id="phoneError" class="error"></span>
```

```
      <button type="submit">Submit</button>
    </form>
  </div>

  <script
src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.6.0/validator.min.js"></script>
  <script src="script.js"></script>
</body>
</html>
```

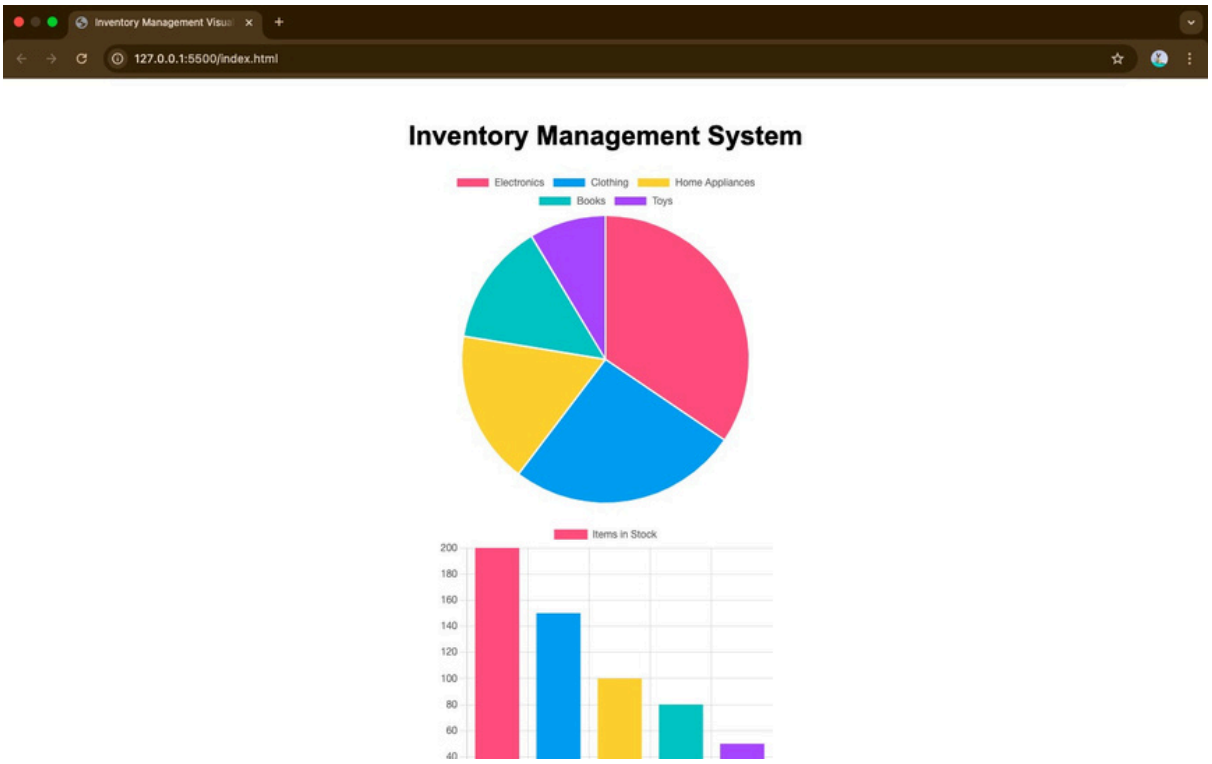
2. JavaScript

```
// Data for the inventory
const inventoryData = {
    labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
    datasets: [
        {
            label: 'Items in Stock',
            data: [200, 150, 100, 80, 50],
            backgroundColor: [
                '#FF6384', '#36A2EB', '#FFCE56', '#4BC0C0', '#9966FF'
            ],
        }
    ]
};

// Creating the Pie Chart
const ctxPie = document.getElementById('pieChart').getContext('2d');
const pieChart = new Chart(ctxPie, {
    type: 'pie',
    data: inventoryData,
    options: {
        responsive: true,
        title: {
            display: true,
            text: 'Inventory Distribution'
        }
    }
});

// Creating the Bar Chart
const ctxBar = document.getElementById('barChart').getContext('2d');
const barChart = new Chart(ctxBar, {
    type: 'bar',
    data: inventoryData,
    options: {
        responsive: true,
        title: {
            display: true,
```

```
            text:'Items in Stock by Category'
        },
        scales: {
            yAxes: [{
                ticks: {
                    beginAtZero: true
                }
            }]
        }
    }
});
```

Output

- Pie Chart showing the distribution of inventory items.



- Bar Chart showing the number of items in each category.

Result

Successfully created pie and bar charts for visualizing inventory data using Chart.js and JavaScript.